Contact

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www.linkedin.com/in/yalinsolak (LinkedIn)
www.chipalizer.com (Other)

Top Skills

Mixed Signal
Semiconductors
IC

Yalin Solak

Founder, R&D Air to Air Combat Drone
San Francisco Bay Area

Summary

Drone R&D Founder/Engineer:

Develop new type of air to air combat capable drones. Counter-UAS and UAS, much of this is in stealth mode, so I can't explain further here. But if you are from DoD or Allied Nations and want to know more, please DM.

Product and Software Engineer:

- Post/silicon device characterization, validation, testing, and comprehensive test data analysis (data-power) including audio codec, general-purpose MCU, LDO, ADC, DAC and clock reference. Software and hardware tools development for testing using VB.NET, Python and Matlab.

- Test Automation software and hardware development design expertise including object-oriented software design, script engine design, interactive user interface design, semiconductor integrated design environments, IC (Integrated Circuit) automated test software, silicon wafer test design, I2C software, instrument integration software, micro-device evaluation board design/software and micro-device firmware implementation.

- Twenty years of software and hardware design experience.

- Software: VB.NET, Python, Matlab, C#, Assembly.

- Hardware: I2C, SPI, GPIB, USB, 1st silicon evalution boards, wafer probes, oscilloscopes, AC/DC power supplies, digital multi-meters, function-generators, DAQs, logic analyzers and temperature chambers. IC device testing and application, including IC products of power management, mixed-signal devices, microcontrollers, MEMs and oscillators.

- Complete system engineering design responsibility including the subsystems, board design, circuit design, IC placement, circuit power optimization and software (firmware) development.

- Board level circuit design with general purpose MCUs, 8-bit micro-controllers, mixed-signal devices, NFC ICs, power management, battery management, energy harvesters, amplifiers, sensors, low power ICs and firmware development.

Experience

Solakair
Founder
February 2025 - Present (6 months)

UAS and Counter Unmanned Aerial System (CUAS) solutions company.

TetraMem - Accelerate The World
Senior Staff Functional Testing Engineer
August 2023 - January 2025 (1 year 6 months)
Fremont, California, United States

Elevatesemi
Senior Software and Product Engineer
July 2022 - March 2023 (9 months)

NextGen Consulting
Project Engineer
September 2019 - June 2022 (2 years 10 months)
Santa Clara, California USA

Microsoft
Senior Electrical Engineer
March 2019 - September 2019 (7 months)
Sunnyvale, California

Work on development of Augmented Reality Glasses

Intel Corporation
Senior Validation Engineer
November 2017 - June 2018 (8 months)
Austin, Texas Area

Qualcomm

Product Engineer

December 2014 - November 2017 (3 years)

Greater San Diego Area

Post-silicon audio codec device characterization, validation, failure analysis, testing, and comprehensive test data analysis (data-power). Provide feedback and testing support to design and system engineers to reach next generation product performance targets.

Chipalizer

Founder and Principal Software Engineer

June 2009 - May 2014 (5 years)

- Designed software development environment that reduced overall micro-device evaluation and validation time by 50%.
- Designed novel semiconductor device software architecture that automated characterization/evaluation software for mixed-signal and power management ICs. Reduced software development from 2-4 months to 1 day.
- Tested prototype evaluation software on LDOs and MEMs product families.

IDT

Software Engineer

March 2012 - February 2013 (1 year)

Ann Arbor, MI

- Improved and maintained bench-top evaluation/validation software environment for crystal-free 100ppm oscillator product lines.
- Designed and implemented IDT-SFC's 48k die wafer probe evaluation software for crystal oscillator IC product lines.
- Designed low level software API that integrated successfully with product evaluation software.

Silicon Laboratories

Design Engineer

May 2006 - April 2009 (3 years)

- Designed bench automated IC characterization, validation and evaluation software development environment that reduced test development schedules by 50%.
- Implemented/led in characterization of six product families, including circuit blocks; ADC, DAC, IDAC, Bandgap, Internal-Osc, External-Osc, Voltage-Regulators, Digital-Logics, POR, Temp-Sensor.

Education

Massachusetts Institute of Technology

Course 6 - Electrical Engineering · (2005 - 2007)

Stevens Institute of Technology

BE, EE · (1997 - 2001)